



Systematic Review

The Efficacy of New Non-Invasive Brain Stimulation in Patients with Chronic Tinnitus Without Specific Treatable Origin: A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

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Abstract

Background/Objectives: Despite the high prevalence (around 4.1–37.2%) and highly debilitating adverse impact, there has been inconclusive evidence regarding the efficacy of treatment for tinnitus management, especially for those patients with tinnitus who do not have a specific or treatable origin. The aim of this updated network meta-analysis (NMA) was to estimate the efficacy and safety of the different non-invasive brain stimulation (NIBS) interventions in tinnitus management in patients with chronic tinnitus without a specific or treatable origin. **Methods:** This NMA included randomized controlled trials (RCTs) of NIBS interventions in patients with chronic tinnitus. The current NMA was conducted using the frequentist model. The primary outcome was a change in tinnitus severity after the NIBS intervention. **Results:** We identified 45 eligible RCTs with a total of 2042 participants. The results of the current NMA showed that both excitatory and inhibitory NIBS interventions exerted significant effects on tinnitus severity, quality of life, or response rate. While several protocols showed a signal, the top-ranked intervention was preliminary and derived from a single, small study. All the NIBS interventions had fair acceptability compared to the controls. **Conclusions:** This NMA highlighted that both excitatory and inhibitory NIBS interventions exerted significant effects on tinnitus severity, quality of life, and/or response rate. Future well-designed RCTs with electroencephalogram applications are needed for replication over the proclamation of efficacy.

Keywords: tinnitus; repetitive transcranial magnetic stimulation; transcranial direct current stimulation; non-invasive brain stimulation; network meta-analysis; systematic review

1. Background

The chronic tinnitus imposed a highly debilitating adverse impact on about 6–25% of affected people [1]. Several diseases might be associated with tinnitus, such as dyslipidaemia, atherosclerosis [2], and lower hemoglobin and platelet levels [3]. Although there have been some consensus guidelines [4,5] or large-scale network meta-analyses (NMAs) [6,7] addressing the efficacy and acceptability of different treatment strategies for tinnitus, the treatment efficacy has still not achieved a satisfactory state for patients with tinnitus. The insufficient treatment efficacy might result from the diverse recognition of the tinnitus physiopathology model. Specifically, there were two distinct theories regarding tinnitus physiopathology. The first one mainly focused on the pathological changes in the hearing system, including the cochlea (or inner ear), vestibulocochlear nerve, and nerve conduction [8–12], and recommended systematic pharmacologic treatments [6]. The other one, based the Jastreboff Neurophysiological Model [13], investigated the limbic system (including the amygdala, anterior cingulate cortex, and insula), frontal cortical areas, and consequent autonomic nervous system alterations [7].

The tinnitus was a complex disease with various systemic diseases involved. For example, in cases of persistent tinnitus without a specific treatable origin, the researchers had noticed a strong association between tinnitus and cardiovascular and metabolic conditions (ex. dyslipidemia and atherosclerosis), which indicated microvascular dysfunction, inflammation, and oxidative stress, in a recent comprehensive analysis by Maihoub et al. [2]. Furthermore, in another important report investigating physiological markers, the authors noticed that hematological parameters (lower hemoglobin and platelet levels) were associated with increased tinnitus severity and chronicity [3]. All these findings pointed out the potential role of oxygen delivery and inflammatory pathways in tinnitus perception, so that tinnitus might not be a simple otologic symptom but a manifestation of systemic

dysregulation, thereby providing a strong rationale for investigating treatments like non-invasive brain stimulation (NIBS) that target central nervous system networks influenced by these peripheral conditions.

In chronic tinnitus without a specific or treatable origin, abnormal hyperactivity in specific brain areas such as the bilateral auditory cortices [14], anterior cingulate cortex [15], and insula [15] has been demonstrated in several clinical trials. The NIBS targeting the indicated brain regions has, therefore, been proposed as a tinnitus management strategy [16]. The NIBS techniques vary widely based on their mechanism and include repetitive transcranial magnetic stimulation (rTMS) [17], rTMS variant theta-burst stimulation (TBS) [18], transcranial direct current stimulation (tDCS) [19], and transcranial random noise stimulation (tRNS) [20]. These different NIBS mechanisms exert different effects on the targeted brain regions [18] and result in different efficacy on tinnitus severity. In our previous large-scale NMA [7], we summarized the currently available NIBS interventions applied to the management of tinnitus without a specific or treatable origin. In our previous NMA, we noticed that NIBS with an inhibitory effect on the auditory cortex was associated with significant efficacy in relieving tinnitus severity [7]. However, in this rising research field, there have been several important and new NIBS strategies for tinnitus management. Most of these studies focused on excitatory NIBS modalities and found a preferable efficacy for tinnitus severity, for example, neuronavigated NIBS [21], tDCS with high current [22], high definition tDCS [23], high-frequency rTMS over dorsomedial prefrontal cortex [24], and assistance of tailor-made notched music training [25]. These new excitatory NIBS strategies showed promising efficacy on tinnitus severity [22,25]. Therefore, to provide up-to-date information and evidence for clinicians, an updated state-of-the-art NMA is required.

Therefore, we conducted an updated NMA of the currently published RCTs to estimate the efficacy and safety of the different NIBS interventions in tinnitus management in patients with tinnitus who do not have a specific or treatable origin.

2. Materials and Methods

2.1. General Guidelines of the Study

The present NMA follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Table S1) and AMSTAR2 (assessing the methodological quality of systematic reviews) guidelines. This study was registered in PROSPERO (CRD42022381048).

2.2. Search Strategy and Selection Criteria

We conducted an updated systematic review using PubMed, Cochrane CENTRAL, ClinicalKey, ProQuest, Embase, ScienceDirect, Web of Science, and ClinicalTrials.gov to identify potentially eligible articles. The initial search date was 2 December 2022, and the final update search was performed on 25 September 2025. No language restrictions were applied. The detailed search strategies and keywords used in each database are listed in Table S2. Additionally, manual searches were performed for potentially eligible articles selected from the reference lists of the review articles and pairwise meta-analyses.

2.3. Inclusion and Exclusion Criteria

Following the rationale of our previous large network meta-analyses (NMAs) [6,7], the PICO applied in the current updated NMA was as follows: (1) participants—patients with chronic tinnitus who do not have a specific or treatable origin; (2) intervention—central NIBS intervention; (3) comparison—sham control or active control or waiting list; and (4) outcomes—changes in tinnitus severity. The definition of “tinnitus without specific or treatable origin” followed the definition of primary tinnitus addressed in the guidelines by

Tunkel et al. (2014) [5]. The non-invasive brain stimulation (NIBS) methods were repetitive transcranial magnetic stimulation (rTMS), transcranial direct current stimulation (tDCS), transcranial random noise stimulation (tRNS), continuous TBS (cTBS), and intermittent TBS (iTBS). To improve the overall quality of the current NMA, we only included peer-reviewed randomized controlled trials (RCTs). Therefore, the inclusion criteria were as follows: (1) RCTs published in peer-reviewed journals; (2) patients with tinnitus who do not have a specific or treatable origin; (3) use of one or more central NIBS interventions as the experimental group; and (4) human trials.

The exclusion criteria were as follows: (1) not a clinical trial, (2) not an RCT, (3) not reporting the target outcomes (defined ahead), and (4) not related to the NIBS methods mentioned earlier. In cases of duplicate use of data (i.e., different articles based on the same sample sources), we included only the article with the largest sample size.

2.4. Data Extraction

Two authors independently screened and extracted relevant data from the included studies. If the original data were not available, the original corresponding authors were electronically contacted via email to request the original data. Discrepancies were adjudicated by a third author. We only extracted data from the NIBS and did not include data on peripheral stimulation.

2.5. Outcome Definition

The primary outcome was the change in tinnitus severity after NIBS intervention in patients with tinnitus without a specific or treatable origin.

The secondary outcomes were changes in quality of life and the response rate related to the NIBS method in patients with tinnitus without a specific or treatable origin. The response rate was defined based on the criteria used in the included studies. Finally, acceptability was defined using the dropout rate, which was defined as the percentage of patients who left early studies for any reason.

2.6. Node Definition

As addressed before, the previous meta-analyses addressing the efficacy of various NIBSs in tinnitus categories (i.e., rTMS versus control or tDCS versus control) lead to wide confidence intervals and significant heterogeneity. Therefore, we decided to specifically separate the various NIBSs based on stimulating sites. As the NIBS method varied widely among the studies, we categorized the methods into five major subgroups: (i) rTMS modalities, which are high-frequency rTMS (HF-rTMS) (>1 Hz) and low-frequency rTMS (LF-rTMS) (≤ 1 Hz); (ii) deep TMS (dTMS) modalities; (iii) TBS modalities, including iTBS and cTBS; (iv) tDCS modalities, which were specifically categorized according to the anodal/cathodal placement position (for example, a-tDCS over F3 or c-tDCS over F3); and (v) a combination of tDCS and other specific interventions, such as tRNS. We further categorized the treatment arms based on the stimulation position according to the electroencephalogram (EEG) brain map following the rationale of classification in our previous NMA of NIBS in different diseases [7].

2.7. Cochrane Risk of Bias Tool

Two independent authors evaluated the risk of bias (inter-rater reliability, kappa = 0.89) for each domain using the Cochrane risk of bias tool. Discrepancies were resolved by consulting a third author.

2.8. Statistical Analysis

NMA was performed using the STATA statistical software (version 16.0; StataCorp LLC, College Station, TX, USA). For continuous variables, we used the standardized mean difference (SMD) with 95% confidence intervals (95% CIs) as the effect size measure. For categorical data, the effect size measure was the odds ratio (OR) with 95% CIs. Although there had not been direct evidence regarding the interpretation of SMD values of THI, we chose the ordinary reports regarding SMD value interpretation in general conditions. Specifically, the SMD value of 0.2 indicated a small effect, 0.5 for medium, and 0.8 for large, respectively [26]. When no events were observed in one of the treatment groups, we applied the continuity correction method by adding 0.5 events to both arms. However, if both the intervention and control arms had no events in a study, this correction procedure was not applied owing to the increased risk of bias. We used the frequentist random effect model for NMA [27]. All comparisons were performed using a two-tailed *t*-test, and a *p*-value of <0.05 was considered statistically significant. Heterogeneity among the included studies was evaluated using the tau value, which is the estimated standard deviation of the effects across the included studies.

Regarding the meta-analysis procedure employed in this study, we used generalized linear mixed models to undertake direct and indirect comparisons [28]. Specifically, indirect comparisons were conducted under the assumption of transitivity, in which the differences between treatments A and B could be calculated from their comparisons with a third treatment C [29]. STATA was used in our NMA with the mvmeta command and self-programmed STATA [27]. The restricted maximum likelihood method was used for the estimation of NMA [30].

To enhance the clinical application of our results, we used the ranking probabilities of treatments compared with the analysis of each outcome to calculate the surface under the cumulative ranking curve (SUCRA). The SUCRA, ranging from 0 to 100, is the percentage of the effect of each treatment relative to an imaginary treatment, which is always the best with absolute certainty [31].

Based on the rationale of the transitivity assumption test used in our previous NMA studies [32], we assessed the variations in the effects of sham or control interventions across different trials to evaluate the validity of the transitivity assumption. Specifically, we computed the sham therapy effect for tDCS sham therapy, rTMS sham therapy, and waiting list by the traditional pairwise meta-analysis using Comprehensive Meta-Analysis (version 3; Biostat, Englewood, NJ, USA) [33]. Finally, we evaluated the potential inconsistencies between the direct and indirect evidence within the network using the loop-specific approach and identified local inconsistencies using the node-splitting method. The design-by-treatment interaction model was used to evaluate global inconsistencies within the entire NMA [34]. We used the comparison-adjusted funnel plots [35] and Egger regression to evaluate potentially small study effects and publication bias within the direct evidence on the comparison between a pair of treatments. In addition, we evaluated the heterogeneity of tau values in the main results of this study. To evaluate the potential placebo effect of sham applications in the RCT with a sham control design, we arranged a subgroup analysis focusing on RCTs with a sham control design.

3. Results

After the initial screening procedure, 142 articles were considered for the full-text review (Figure 1). However, 97 were excluded for various reasons (Table S3). Finally, after adding 13 new RCTs to our previous NMA, 45 articles were included in the current study (Table S4). Figure 2 depicts the entire geometric distribution of the treatment arms.

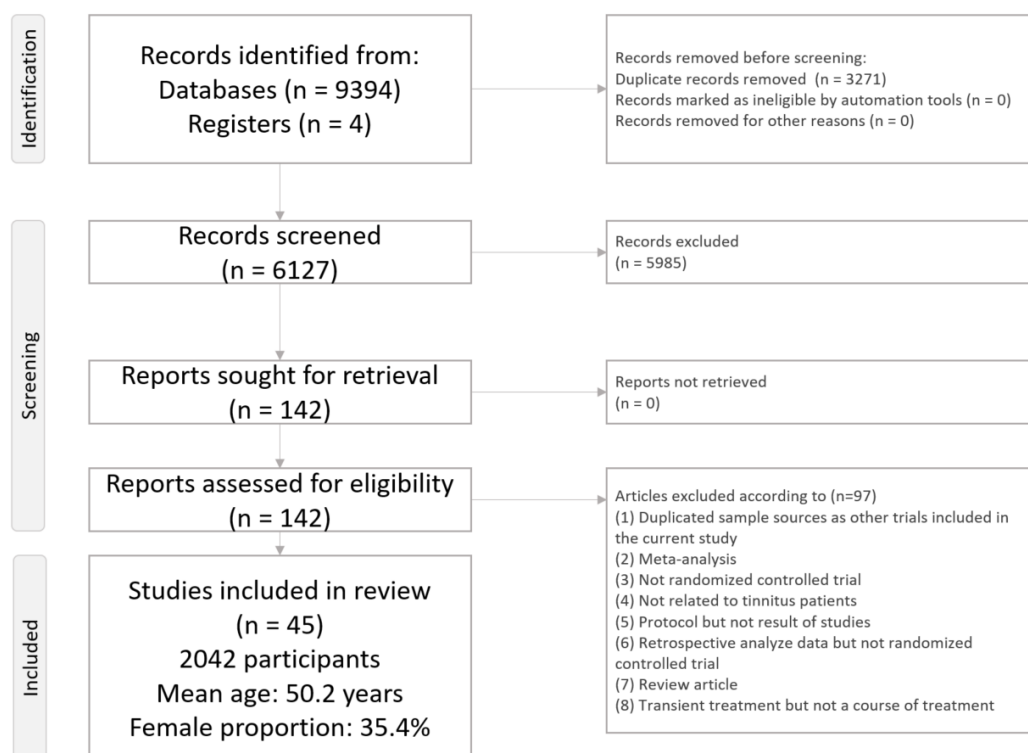


Figure 1. Flowchart of the current network meta-analysis.

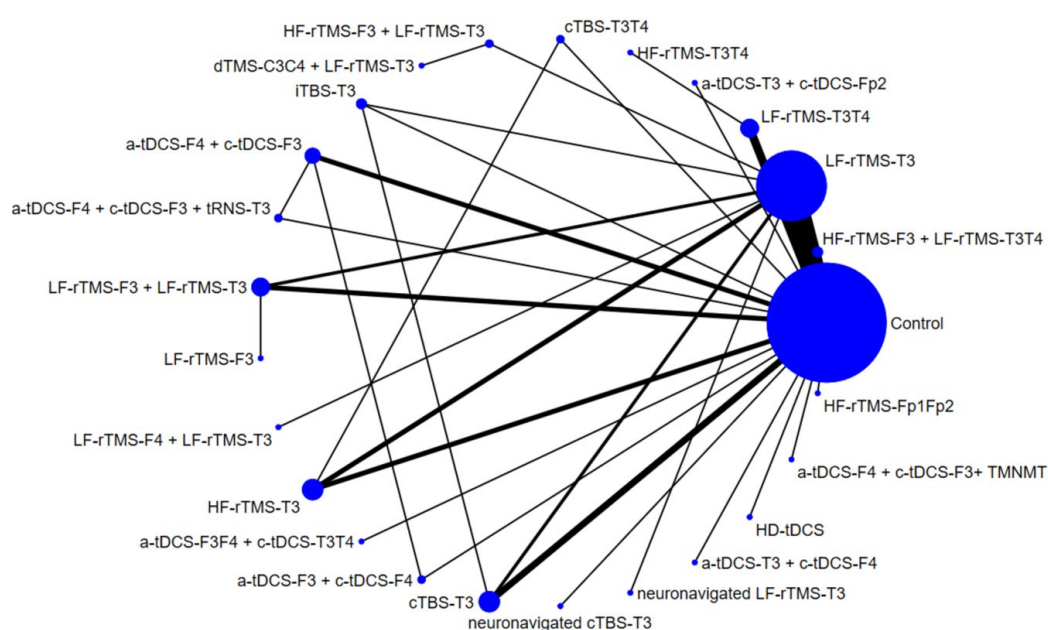


Figure 2. The network structure of changes in the severity of tinnitus. Overall structure of the network meta-analysis. The lines between nodes represent direct comparisons in various trials, and the size of each circle is proportional to the number of participants in each specific treatment. The thickness of the lines is proportional to the number of trials connected to the network.

Figure 1 depicts the entire flowchart of the current network meta-analysis.

Figure 2 depicts the overall network structure of the current network meta-analysis of the changes in tinnitus severity. The lines between nodes represent direct comparisons in various trials, and the size of each circle is proportional to the size of the population involved in each specific treatment. The thickness of the lines is proportional to the number of trials connected to the network.

3.1. Characteristics of the Included Studies

A total of 2042 participants were included in the study. The mean age of the participants was 50.2 years (range = 40.0–60.55 years), and the mean proportion of females was 35.4% (range = 0.0–79.7%). The mean duration of NIBS treatment was 15.1 weeks (range, 1–54 weeks). The baseline characteristics of the participants are summarized in Table S4. The definition of response varied between the recruited studies, with patient self-rated global impression, tinnitus handicap inventory (THI) score reduction > 7, 10, or 20 points, THI score reduction > 20% of the baseline THI, and tinnitus questionnaire (TQ) score reduction > 5 or 10 points.

3.2. Primary Outcome: Change in Tinnitus Severity

The NMA revealed that only the a-tDCS-T3 + c-tDCS-F4 (SMD = -2.55 , 95% CIs = -3.86 to -1.25), a-tDCS-F4 + c-tDCS-F3 + tRNS-T3 (SMD = -1.97 , 95% CIs = -2.98 to -0.96), a-tDCS-F4 + c-tDCS-F3 + TMNMT (SMD = -1.67 , 95% CIs = -2.80 to -0.54), HF-rTMS-Fp1Fp2 (SMD = -1.34 , 95% CIs = -2.40 to -0.29), cTBS-T3T4 (SMD = -0.72 , 95% CIs = -1.41 to -0.04), HF-rTMS-F3 + LF-rTMS-T3T4 (SMD = -0.71 , 95% CIs = -1.32 to -0.10), and LF-rTMS-T3 (SMD = -0.30 , 95% CIs = -0.55 to -0.06) were associated with significant improvement in the severity of tinnitus compared to the controls (Tables 1 and 2, and Figure 3). The main results would not be changed after removing single-study nodes. According to the SUCRA, a-tDCS-T3 + c-tDCS-F4 was associated with the largest improvement, followed by a-tDCS-F4 + c-tDCS-F3 + tRNS-T3, and a-tDCS-F4 + c-tDCS-F3 + TMNMT (Table S5A). The transitivity assumption test was verified using the interaction test. There was no significant difference in placebo effects between the rTMS sham therapy, tDCS sham therapy, and waiting list ($p = 0.355$; Figure S2A). However, a significant placebo effect was detected in the rTMS sham controls (SMD = -0.152 , 95% CIs: -0.300 to -0.004 , $p = 0.045$).

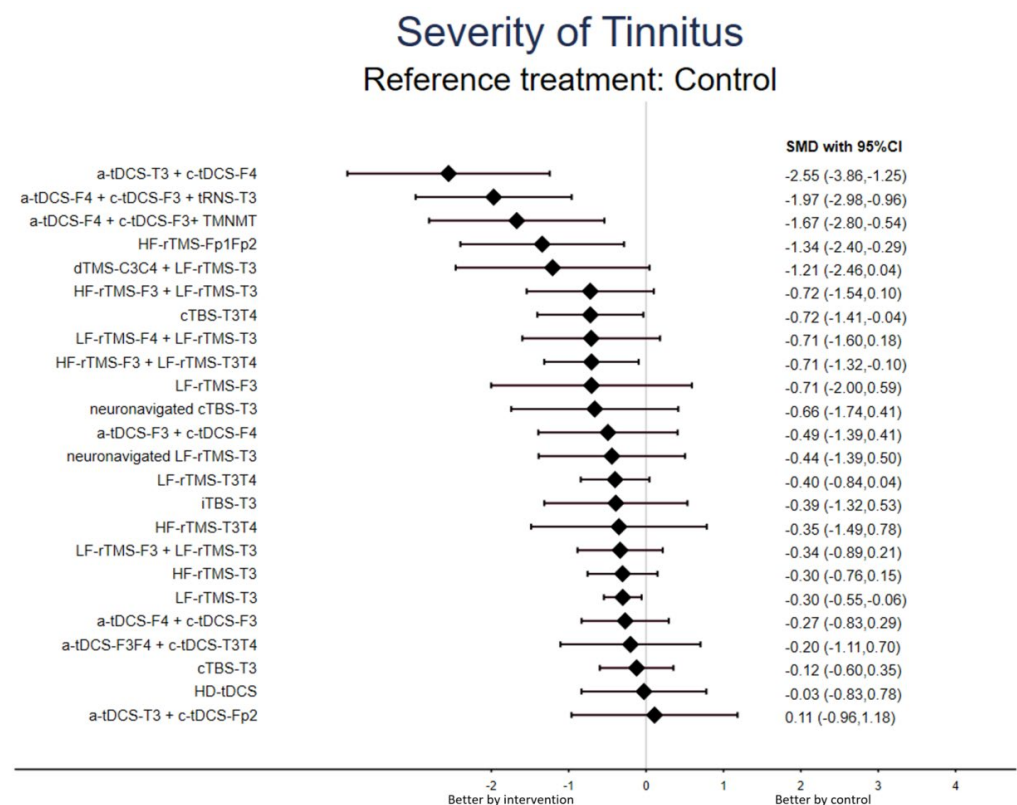


Figure 3. Forest plot of the changes in the severity of tinnitus.

Table 1. League table of the improvement in the severity of tinnitus.

a-tDCS-T3 + c-tDCS-F4																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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Table 1. *Cont.*

* −2.15 (−3.53, −0.77)	* −1.57 (−2.67, −0.47)	* −1.27 (−2.49, −0.05)	−0.94 (−2.09, 0.20)	−0.81 (−2.14, 0.52)	−0.31 (−1.06, 0.45)	−0.32 (−1.25, 0.61)	−0.32 (−1.14, 0.50)	−0.31 (−1.30, 0.69)	−0.30 (−1.67, 1.07)	−0.26 (−1.43, 0.90)	−0.09 (−1.09, 0.91)	−0.04 (−1.09, 1.00)	0.01 (−1.02, 1.04)	LF- rTMS- T3T4	−0.05 (−0.85, 0.75)														* (−0.39 (−0.72, −0.07)
* −2.20 (−3.93, −0.47)	* −1.62 (−3.14, −0.10)	−1.32 (−2.92, 0.28)	−0.99 (−2.54, 0.56)	−0.86 (−2.55, 0.83)	−0.36 (−1.64, 0.93)	−0.37 (−1.77, 1.03)	−0.37 (−1.69, 0.96)	−0.36 (−1.80, 1.08)	−0.35 (−2.08, 1.37)	−0.31 (−1.88, 1.25)	−0.14 (−1.59, 1.31)	−0.09 (−1.57, 1.39)	−0.04 (−1.50, 1.42)	−0.05 (−1.09, 0.98)	HF- rTMS- T3T4														
* −2.22 (−3.64, −0.80)	* −1.63 (−2.78, −0.49)	* −1.34 (−2.60, −0.08)	−1.01 (−2.20, 0.18)	−0.87 (−2.23, 0.48)	−0.37 (−1.18, 0.44)	−0.39 (−1.35, 0.58)	−0.38 (−1.26, 0.49)	−0.37 (−1.40, 0.65)	−0.37 (−1.54, 0.80)	−0.33 (−1.54, 0.88)	−0.16 (−1.21, 0.90)	−0.11 (−1.18, 0.97)		−0.06 (−1.12, 1.01)	−0.07 (−0.77, 0.64)	−0.02 (−1.28, 1.25)	LF- rTMS- F3 + LF- rTMS- T3			−0.26 (−0.83, 0.31)								−0.23 (−0.68, 0.22)	
* −2.25 (−3.63, −0.87)	* −1.67 (−2.77, −0.56)	* −1.37 (−2.59, −0.15)	−1.04 (−2.19, 0.11)	−0.90 (−2.21, 0.41)	−0.40 (−1.15, 0.35)	−0.42 (−1.33, 0.49)	−0.42 (−1.09, 0.26)	−0.41 (−1.38, 0.57)	−0.40 (−1.77, 0.97)	−0.36 (−1.53, 0.81)	−0.19 (−1.19, 0.82)	−0.14 (−1.16, 0.88)	−0.09 (−1.10, 0.93)	−0.10 (−0.73, 0.54)	−0.05 (−1.27, 1.17)	−0.03 (−0.73, 0.67)	HF- rTMS- T3			−0.10 (−0.63, 0.43)								* (−1.15, −0.05)	
* −2.25 (−3.58, −0.92)	* −1.67 (−2.70, −0.63)	* −1.37 (−2.53, −0.21)	−1.04 (−2.13, 0.04)	−0.91 (−2.13, 0.32)	−0.40 (−1.03, 0.22)	−0.42 (−1.21, 0.36)	−0.42 (−1.13, 0.29)	−0.41 (−1.26, 0.45)	−0.40 (−1.71, 0.90)	−0.36 (−1.47, 0.74)	−0.19 (−1.12, 0.74)	−0.14 (−1.05, 0.77)	−0.09 (−1.01, 0.84)	−0.10 (−0.60, 0.41)	−0.05 (−1.21, 1.11)	−0.03 (−0.60, 0.53)	−0.00 (−0.46, 0.45)	LF- rTMS- T3							* −0.74 (−1.30, −0.18)			* (−0.32 (−0.54, −0.09)	
* −2.28 (−3.71, −0.86)	* −1.70 (−2.72, −0.68)	* −1.40 (−2.67, −0.14)	−1.07 (−2.27, 0.13)	−0.94 (−2.31, 0.44)	−0.44 (−1.27, 0.39)	−0.45 (−1.45, 0.55)	−0.45 (−1.34, 0.44)	−0.44 (−1.49, 0.61)	−0.43 (−1.85, 0.98)	−0.39 (−1.61, 0.82)	−0.22 (−1.12, 0.68)	−0.17 (−1.27, 0.93)	−0.12 (−1.20, 0.96)	−0.13 (−0.85, 0.59)	−0.08 (−1.35, 1.19)	−0.07 (−0.85, 0.72)	−0.03 (−0.76, 0.69)	−0.03 (−0.65, 0.58)	a-tDCS- F4 + c-tDCS- F3								−0.29 (−0.70, 0.12)		
* −2.35 (−3.94, −0.76)	* −1.77 (−3.12, −0.41)	* −1.47 (−2.92, −0.02)	−1.14 (−2.53, 0.25)	−1.01 (−2.55, 0.54)	−0.50 (−1.60, 0.59)	−0.52 (−1.74, 0.70)	−0.52 (−1.65, 0.62)	−0.51 (−1.78, 0.76)	−0.50 (−2.08, 1.08)	−0.46 (−1.87, 0.95)	−0.29 (−1.57, 0.98)	−0.24 (−1.55, 1.07)	−0.19 (−1.48, 1.10)	−0.20 (−1.21, 0.81)	−0.15 (−1.60, 1.30)	−0.13 (−1.19, 0.93)	−0.10 (−1.11, 0.91)	−0.10 (−1.04, 0.84)	−0.07 (−1.13, 1.00)	a-tDCS- F3F4 + c-tDCS- T3T4							−0.20 (−0.81, 0.41)		
* −2.43 (−3.82, −1.04)	* −1.85 (−2.96, −0.73)	* −1.55 (−2.78, −0.32)	* −1.22 (−2.38, −0.06)	−1.09 (−2.41, 0.24)	−0.58 (−1.35, 0.18)	−0.60 (−1.53, 0.33)	−0.60 (−1.43, 0.23)	−0.59 (−1.57, 0.40)	−0.58 (−1.96, 0.79)	−0.54 (−1.72, 0.64)	−0.37 (−1.39, 0.65)	−0.32 (−1.36, 0.72)	−0.27 (−1.22, 0.69)	−0.28 (−0.93, 0.37)	−0.23 (−1.46, 1.00)	−0.21 (−0.93, 0.50)	−0.18 (−0.82, 0.46)	−0.18 (−0.67, 0.31)	−0.15 (−0.89, 0.59)	−0.08 (−1.10, 0.94)	cTBS-T3						−0.52 (−1.22, 0.19)		
* −2.52 (−4.06, −0.99)	* −1.94 (−3.23, −0.65)	* −1.64 (−3.03, −0.25)	−1.31 (−2.67, 0.02)	−1.18 (−2.63, 0.31)	−0.68 (−1.69, 0.33)	−0.69 (−1.85, 0.46)	−0.69 (−1.75, 0.37)	−0.68 (−1.88, 0.52)	−0.68 (−2.20, 0.85)	−0.64 (−1.98, 0.71)	−0.46 (−1.67, 0.74)	−0.41 (−1.66, 0.83)	−0.36 (−1.59, 0.86)	−0.37 (−1.29, 0.55)	−0.32 (−1.71, 1.07)	−0.31 (−1.28, 0.67)	−0.28 (−1.29, 0.65)	−0.27 (−1.12, 0.57)	−0.24 (−1.23, 0.74)	−0.17 (−1.39, 1.04)	−0.09 (−1.03, 0.84)	HD- tDCS					−0.03 (−0.48, 0.42)		
* −2.66 (−4.35, −0.97)	* −2.08 (−3.55, −0.61)	* −1.78 (−3.34, −0.22)	−1.45 (−2.96, 0.05)	−1.32 (−2.97, 0.33)	−0.82 (−2.05, 0.42)	−0.83 (−2.18, 0.52)	−0.83 (−2.10, 0.44)	−0.82 (−2.21, 0.58)	−0.81 (−2.50, 0.87)	−0.77 (−2.29, 0.75)	−0.60 (−2.00, 0.80)	−0.55 (−1.98, 0.88)	−0.50 (−1.92, 0.92)	−0.51 (−1.67, 0.65)	−0.46 (−2.02, 1.10)	−0.44 (−1.65, 0.76)	−0.41 (−1.58, 0.75)	−0.41 (−1.51, 0.69)	−0.38 (−1.59, 0.83)	−0.31 (−1.71, 1.09)	−0.23 (−1.40, 0.94)	−0.14 (−1.48, 1.20)					0.11 (−0.73, 0.95)		
* −2.55 (−3.86, −1.25)	* −1.97 (−2.98, −0.96)	* −1.67 (−2.80, −0.54)	* −1.34 (−2.40, 0.04)	−1.21 (−2.46, 0.04)	* −0.71 (−1.32, −0.10)	−0.72 (−1.54, 0.10)	* −0.72 (−1.41, −0.04)	−0.71 (−1.60, 0.18)	−0.71 (−2.00, 0.59)	−0.66 (−1.74, 0.41)	−0.49 (−1.39, 0.41)	−0.44 (−1.39, 0.50)	−0.39 (−1.32, 0.53)	−0.40 (−0.84, 0.04)	−0.35 (−1.49, 0.78)	−0.34 (−0.89, 0.21)	−0.30 (−0.76, 0.15)	* −0.30 (−0.55, −0.06)	−0.27 (−0.83, 0.29)	−0.20 (−1.11, 0.70)	−0.12 (−0.60, 0.35)	−0.03 (−0.83, 0.78)	0.11 (−0.96, 1.18)				Sham		

This league table presents both pairwise (upper-right portion) and network (lower-left portion) meta-analysis results of improvement in severity of tinnitus in patients with tinnitus. We ordered the interventions according to the rank of efficacy of tinnitus improvement. The outcomes are expressed as standardized mean difference (SMD) (95% confidence intervals). For the pairwise meta-analyses, an SMD of less than 0 indicates that the treatment specified in the row achieved more improvement than that specified in the column. For the network meta-analysis (NMA), an SMD of less than 0 indicates that the treatment specified in the column got more improvement than that specified in the row. Bold results marked with * indicate statistical significance.

Table 2. Confidence of evidence regarding the protocols/interventions investigated.

Confidence of Evidence	Protocols/Interventions	Standardized Mean Difference	95% Confidence Intervals
Consistent evidence	cTBS-T3T4	−0.72	−1.41 to −0.04
	HF-rTMS-F3 + LF-rTMS-T3T4	−0.71	−1.32 to −0.10
	LF-rTMS-T3	−0.30	−0.55 to −0.06
Single-study nodes	a-tDCS-T3 + c-tDCS-F4	−2.55	−3.86 to −1.25
	a-tDCS-F4 + c-tDCS-F3 + tRNS-T3	−1.97	−2.98 to −0.96
	a-tDCS-F4 + c-tDCS-F3 + TMNMT	−1.67	−2.80 to −0.54
	HF-rTMS-Fp1Fp2	−1.34	−2.40 to −0.29

This table presents the evidence confidence. The protocols/interventions listed in “Consistent evidence” consisted of at least 2 trials and with relatively high precision, whereas the protocols/interventions listed in “Single-study nodes” consisted of only 1 trial and with relatively insufficient precision.

Figure 3 indicates that when the effect size is less than zero, the specified treatment was associated with greater improvement in the severity of tinnitus than controls.

3.3. Primary Outcome: Change in Tinnitus Severity in the Subgroup of Sham-Controlled Trials

In the subgroup analysis of sham-controlled RCTs, the main result of the NMA did not change. To be specific, only the a-tDCS-T3 + c-tDCS-F4 (SMD = −2.55, 95% CIs = −3.79 to −1.32), a-tDCS-F4 + c-tDCS-F3 + TMNMT (SMD = −1.67, 95% CIs = −2.72 to −0.63), HF-rTMS-Fp1Fp2 (SMD = −1.34, 95% CIs = −2.31 to −0.38), HF-rTMS-F3 + LF-rTMS-T3T4 (SMD = −0.71, 95% CIs = −1.25 to −0.17), HF-rTMS-T3 (SMD = −0.50, 95% CIs = −0.92 to −0.07), LF-rTMS-T3T4 (SMD = −0.40, 95% CIs = −0.79 to −0.01), and LF-rTMS-T3 (SMD = −0.29, 95% CIs = −0.51 to −0.07) were associated with significant improvement in the severity of tinnitus compared to the controls (Table S6A, Figures S1A and S2B). According to the SUCRA, a-tDCS-T3 + c-tDCS-F4 was associated with the largest improvement, followed by a-tDCS-F4 + c-tDCS-F3 + TMNMT, and HF-rTMS-Fp1Fp2 (Table S5B).

3.4. Secondary Outcome: Change in Quality of Life

The NMA revealed that only the a-tDCS-T3 + c-tDCS-F4 (SMD = −2.45, 95% CIs = −3.63 to −1.28), a-tDCS-F4 + c-tDCS-F3 + TMNMT (SMD = −1.41, 95% CIs = −2.38 to −0.44), a-tDCS-F4 + c-tDCS-F3 + tRNS-T3 (SMD = −1.28, 95% CIs = −2.11 to −0.46), HF-rTMS-Fp1Fp2 (SMD = −1.01, 95% CIs = −1.90 to −0.13), LF-rTMS-F3 + LF-rTMS-T3 (SMD = −0.83, 95% CIs = −1.41 to −0.25), LF-rTMS-T3T4 (SMD = −0.51, 95% CIs = −0.87 to −0.16), HF-rTMS-T3 (SMD = −0.44, 95% CIs = −0.84 to −0.05), and LF-rTMS-T3 (SMD = −0.34, 95% CIs = −0.56 to −0.11) were associated with significant improvements in the quality of life in patients with tinnitus compared to the controls (Table S6B, Figures S1B and S2C). According to the SUCRA, a-tDCS-T3 + c-tDCS-F4 was associated with the largest improvement in quality of life, followed by a-tDCS-F4 + c-tDCS-F3 + TMNMT, and a-tDCS-F4 + c-tDCS-F3 + tRNS-T3 (Table S5C).

3.5. Secondary Outcome: Response Rate

The NMA revealed that only HF-rTMS-T3 (OR = 6.68, 95% CIs = 1.74 to 25.58), LF-rTMS-F3 + LF-rTMS-T3 (OR = 5.16, 95% CIs = 1.88 to 14.16), and LF-rTMS-T3T4 (OR = 4.59, 95% CIs = 1.24 to 17.00) were associated with significantly better response rates than the controls (Table S6C, Figures S1C and S2D). According to SUCRA, HF-rTMS-Fp1Fp2 (OR = 15.19, 95% CIs = 0.69 to 333.18 compared to controls) was associated with the highest

response rate, followed by HF-rTMS-T3. However, the result of HF-rTMS-Fp1Fp2, which had wide confidence intervals, indicated an extreme imprecision so that the result might be less reliable and needs future research to replicate it (Table S5D).

3.6. Acceptability in the Dropout Rate

In the NMA, none of the investigated NIBSs were associated with significantly different dropout rates compared with the controls (Tables S5E and S6D, Figures S1D and S2E).

3.7. Risk of Bias, Funnel Plot, Heterogeneity, and Publication Bias

We found that 61.6% (194/315 items), 30.8% (97/315 items), and 7.6% (24/315 items) of the included studies had low, unclear, and high risks of bias, respectively. Unclear reporting of allocation concealment or blindness of the studies further contributed to the risk of bias (Figure S3A,B).

Funnel plots of publication bias (Figure S4A–J) revealed general symmetry, and the Egger's test results indicated no significant publication bias among the articles included in the NMA. In general, the NMA did not demonstrate inconsistencies in terms of either local inconsistencies, as assessed using the loop-specific approach and node-splitting method, or global inconsistencies, as determined using the design-by-treatment method (Tables S7 and S8). In addition, according to the I^2 and τ^2 in the heterogeneity test, there was no significant heterogeneity in the main results of this study (detailed I^2 and τ^2 value please see Table S9). The GRADE ratings revealed the overall quality of evidence to be low to medium (Table S10).

4. Discussion

In contrast to our previous NMA, which mainly addressed inhibitory NIBS efficacy on tinnitus management [7], this updated NMA found that both excitatory (i.e., high current a-tDCS-T3 + c-tDCS-F4, high current a-tDCS-F4 + c-tDCS-F3 + TMNMT, HF-rTMS-Fp1Fp2, and high current a-tDCS-F4 + c-tDCS-F3 + tRNS-T3) and inhibitory NIBS (i.e., cTBS-T3T4, HF-rTMS-F3 + LF-rTMS-T3T4, and LF-rTMS-T3) exerted significant treatment effects on tinnitus severity, quality of life, and the response rate. Further, a-tDCS-T3 + c-tDCS-F4 was associated with the largest improvement in both tinnitus severity and quality of life among all NIBS treatments investigated. However, before applying our results in clinical practice, the clinicians should pay special attention due to the predominance of single-study nodes, small sample sizes, and heterogeneity in protocols and outcome measures among the recruited RCTs in this NMA. To date, tDCS has been found to have divergent brain-acting effects depending on its polarity. Specifically, the excitatory or inhibitory effect of tDCS might rely on its current-dependent effect but not polarization-dependent effect [36–38]. The evidence of the efficacy of a-tDCS-T3 + c-tDCS-F4 in the current NMA mainly came from one RCT by Souza et al. (2020) [22], in which the intensity of the current was 2 mA, so it fell within the excitatory effect range. In that previous RCT, the authors also investigated EEG before and after tDCS application and found that tDCS application modulated the theta band in the limbic system and beta band in the temporal regions, which were associated with the tinnitus component (i.e., attention and perception). However, because the efficacy of a-tDCS-T3 + c-tDCS-F4 mainly came from one RCT [22], the evidence of a-tDCS-T3 + c-tDCS-F4 in tinnitus management remains preliminary, so clinicians should be especially careful when applying this finding in their clinical practice. Therefore, the current NMA cannot definitively recommend one protocol over another for clinical practice due to this evidence's fragility.

Another remarkable finding of the current NMA was that among these NIBS interventions with statistically significant efficacy, both excitatory (i.e., high current

a-tDCS-T3 + c-tDCS-F4, high current a-tDCS-F4 + c-tDCS-F3 + TMNMT, HF-rTMS-Fp1Fp2, and high current a-tDCS-F4 + c-tDCS-F3 + tRNS-T3) and inhibitory NIBS (i.e., cTBS-T3T4, HF-rTMS-F3 + LF-rTMS-T3T4, and LF-rTMS-T3) exerted significant treatment effects on tinnitus severity, quality of life, and the response rate. These conflicting findings might not be consistent with the previous findings regarding the hyper-excitability in the cortical regions in patients with tinnitus. Rather, there has recently been another newly developed hypothesis regarding the varied efficacy of NIBS [39]. The “neural noise hypothesis” suggested that the after-effect of NIBS might mainly rely on the overall altered activity of the dopaminergic, GABAergic, glutamatergic, and serotonergic synapses through the network axon projection connection within the whole brain [40]. Central gain was another important physiopathology based on the neural noise hypothesis, which signals were the main target of brain stimulation [7]. As addressed in Qi’s report [41], abnormally enhanced bidirectional EEG activities across multiple timescales and frequency bands had been found in tinnitus patients. Similar findings could also be found in the report by Shen and colleagues [42], in which the authors found a complex reorganization in the brain EEG activities in idiopathic tinnitus patients. Therefore, the improvement in tinnitus severity, quality of life, or response rate might not be simply determined by the depolarization or hyperpolarization in different cortices related to the after-effect of specific NIBS on specific brain regions. Rather, we should consider the whole brain network [40] and acknowledge that excitatory or inhibitory NIBS interventions could both contribute to a rectified effect on the whole brain function so that it might achieve a more steady state after the NIBS interventions [22]. The rectification theory was mainly based on the desynchronized neural conduction in the brain region and led to tinnitus sound formation [43]. The theory stated that rectifying the desynchronized signals in the brain would theoretically manage (or reduce) tinnitus severity [44]. Therefore, needs for further multi-center, well-designed RCTs with EEG applications was still strong to replicate the findings of promising single-study protocols (like a-tDCS-T3 + c-tDCS-F4). (The discussion regarding the placebo effect was depicted in Table S11).

Several potential limitations should be considered in the current NMA. First, this NMA may have been underpowered due to the heterogeneity of the participants (e.g., comorbidities, physiological measures, baseline severity of tinnitus, duration of tinnitus onset, commercial machine used in each study, and follow-up duration), variety in the definition of response, and variety in tinnitus severity or quality of life rating scales. Those varieties might not be simply dealt with statistics measurement and might impose potential fundamental bias on the result of this NMA. Since there had not been a consistent consensus on the definition of chronic tinnitus across countries [45], the wide variation in tinnitus duration might imply potential bias for the result of this study. Although there was no statistically significant heterogeneity or inconsistency, some differences were observed between the included RCTs, which could be attributed to other unknown factors. Despite our effort to include a large number of RCTs (i.e., 45 RCTs), the variability in stimulation parameters, patient demographics, and tinnitus characteristics across studies might limit the interpretation of this study. Specifically, these factors significantly influence the pooling and interpretation of results. Second, although most RCTs included a sham control in their study design, the blinding of those RCTs may not have been complete because of the limitations of the commercial machine employed. Third, because there have been some studies with small sample sizes and short follow-up duration, there might be some potential small outliers within the included RCTs. The efficacy during long term follow-up should be interpreted with caution. To investigate this issue, we evaluated the funnel plot and found no significant outliers within the included RCTs. Further, the main results of our study remained similar after removing single-study nodes. However, the efficacy of

a-tDCS-T3 + c-tDCS-F4 mainly came from one RCT with small sample sizes ($n = 27$) [22]. Specifically, the SUCRA, which ranked treatments based on the position within the network comparison, might be adversely affected by a single trial with extensive efficacy values. Therefore, the evidence of this treatment arm should be interpreted with caution. Another potential limitation of this study should be addressed. Specifically, the wide variety in rating scales of tinnitus severity had been applied among the recruited RCTs. Although the cut point of “clinically meaningful changes in THI” had been defined [46], this was not applicable to most other rating scales. Therefore, the statistical significance might not always reflect the clinical significance. Although we used response rate to reflect clinically relevant improvement, the various definitions of response rate might still impose potential bias on the results of our study. In addition, to reduce potential heterogeneity among the treatment arms, we only included NIBS in the current study and excluded other treatment options, such as pharmacologic treatments and auditory training/sound therapy. Although this selection would increase homogeneity and consistency, it would make us miss the comparison with other treatment options. Furthermore, although we extensively divide various NIBSs into separate treatment arms to provide the most comprehensive information, this process would lead to the consequently small sample sizes in some treatment arms. Finally, the concerns of the transitivity assumption among this NMA might need to be addressed. However, this might not be required because the transitivity assumption test in the current NMA revealed no significantly different placebo effects between the rTMS sham therapy, tDCS sham therapy, and waiting list. Additionally, some RCTs lacked complete documentation of ethical approval and consent. Therefore, future research with comprehensive reporting of ethical approval and consent procedures to enhance transparency should be warranted.

5. Conclusions

The excitatory and inhibitory NIBS both exerted significant treatment effects on tinnitus severity, quality of life, and the response rate. All the NIBS interventions had fair acceptability compared to the controls. Future well-designed RCTs with EEG applications are warranted to explore the role of NIBS in tinnitus management.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/ohbm7010007/s1>. Figure S1: Network structure; Figure S2: Transitivity assumption; Figure S3: Risk of bias; Figure S4: Funnel plot; Table S1: PRISMA checklist; Table S2: Keyword and search result; Table S3: Excluded studies; Table S4: Included studies; Table S5: SUCRA; Table S6: League table; Table S7: Inconsistency; Table S8: Between study standard deviation; Table S9: Heterogeneity; Table S10: GRADE; Table S11: Supplementary discussion. References [17–25,44,45,47–180] are cited in supplementary document.

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Abbreviations

95% CI	95% confidence interval
a-tDCS-F3 + c-tDCS-F4	anode tDCS over left DLPFC plus cathode tDCS over right DLPFC
a-tDCS-F3F4 + c-tDCS-T3T4	anode tDCS over bilateral DLPFC plus cathode tDCS over bilateral auditory cortex
a-tDCS-F4 + c-tDCS-F3 + tRNS-T3	anode tDCS over right DLPFC plus cathode tDCS over left DLPFC plus tRNS over left auditory cortex
a-tDCS-F4 + c-tDCS-F3	anode tDCS over right DLPFC plus cathode tDCS over left DLPFC
a-tDCS-F4 + c-tDCS-F3+ TMNMT	anode tDCS over right DLPFC plus cathode tDCS over left DLPFC plus tailor-made notched music training
a-tDCS-T3 + c-tDCS-F4	anode tDCS over left auditory cortex plus cathode tDCS over right DLPFC
a-tDCS-T3 + c-tDCS-Fp2	anode tDCS over left auditory cortex plus cathode tDCS over right supraorbital area
cTBS	continuous theta-burst stimulation
cTBS-T3	cTBS over left auditory cortex
cTBS-T3T4	cTBS over bilateral auditory cortex
dTMS	deep transcranial magnetic stimulation
dTMS-C3C4 + LF-rTMS-T3	deep TMS over bilateral anterior-cingulate-cortex plus low-frequency rTMS over left auditory cortex
EEG	electroencephalogram
HD-tDCS	high-definition tDCS
HF-rTMS-F3 + LF-rTMS-T3	high-frequency rTMS over left DLPFC plus low-frequency rTMS over left auditory cortex
HF-rTMS-F3 + LF-rTMS-T3T4	high-frequency rTMS over left DLPFC plus low-frequency rTMS over bilateral auditory cortex
HF-rTMS-Fp1Fp2	high-frequency rTMS over bilateral DMPFC
HF-rTMS-T3	high-frequency rTMS over left auditory cortex
HF-rTMS-T3T4	high-frequency rTMS over bilateral auditory cortex
iTBS	intermittent theta-burst stimulation
iTBS-T3	iTBS over left auditory cortex
LF-rTMS-F3 + LF-rTMS-T3	low-frequency rTMS over left DLPFC plus low-frequency rTMS over left auditory cortex
LF-rTMS-F3	low-frequency rTMS over left DLPFC
LF-rTMS-F4 + LF-rTMS-T3	low-frequency rTMS over right DLPFC plus low-frequency rTMS over left auditory cortex
LF-rTMS-T3	low-frequency rTMS over left auditory cortex
LF-rTMS-T3T4	low-frequency rTMS over bilateral auditory cortex
neuronavigated cTBS-T3	neuronavigated cTBS over left auditory cortex
neuronavigated LF-rTMS-T3	neuronavigated low-frequency rTMS over left auditory cortex
NIBS	non-invasive brain stimulation
NMA	network meta-analysis
OR	odds ratio
RCT	randomized controlled trial
rTMS	repetitive transcranial magnetic stimulation
Sham	sham control

SMD	standardized mean difference
SUCRA	surface under the cumulative ranking curve
tDCS	transcranial direct current stimulation
TMNMT	tailor-made notched music training
tRNS	transcranial random noise stimulation

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